

BUECKEBURG, Germany

WMO: 103350

Lat: **52.2817N**Lon: **9.0889E**Elev: **77**StdP: **100.41**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.9	-6.3	-12.0	1.3	-7.0	-9.3	1.7	-5.3	13.3	7.6	12.2	7.7	2.7	80	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	30.8	19.9	28.6	19.1	26.4	18.3	20.9	28.3	19.9	26.6	19.0	25.1	3.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	13.4	23.2	17.6	12.7	22.3	16.8	12.1	21.6	60.6	28.4	57.1	26.6	54.2	25.1	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.3	8.2	-11.6	34.2	4.1	1.6	-14.5	35.4	-16.9	36.4	-19.2	37.3	-22.2	38.5		
(5)			-11.9	23.0	4.0	1.3	-14.8	23.9	-17.1	24.7	-19.3	25.4	-22.2	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	2.3	3.2	5.7	9.8	13.6	16.7	19.0	18.9	15.2	11.0	6.5	3.5
	DBStd	7.06	4.70	4.44	3.91	3.94	3.66	3.43	3.38	3.20	3.16	3.67	3.73	4.42
	HDD10.0	983	239	192	140	52	8	0	0	0	1	33	115	204
	HDD18.3	3003	497	423	391	256	152	72	32	31	105	229	356	460
	CDD10.0	1164	1	2	7	46	121	202	280	275	156	63	9	2
	CDD18.3	142	0	0	0	1	7	23	53	48	10	1	0	0
	CDH23.3	1317	0	0	0	8	68	218	514	431	76	2	0	0
	CDH26.7	409	0	0	0	0	11	56	178	148	16	0	0	0
Wind	WSAvg	3.7	4.7	4.4	4.2	3.5	3.2	3.1	3.1	3.1	3.3	3.8	4.0	4.4
Precipitation	PrecAvg	748	66	51	48	46	62	63	77	78	64	63	64	65
	PrecMax	1023	119	110	110	95	150	137	180	202	142	187	126	161
	PrecMin	549	11	14	8	6	22	23	28	19	16	16	5	21
	PrecStd	116	30	26	27	22	33	25	34	38	30	32	30	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	14.3	18.9	24.4	28.3	31.1	33.7	33.2	29.0	22.2	16.2	13.6
		MCWB	9.6	9.2	11.2	15.1	18.6	20.9	20.3	20.4	18.8	16.6	12.7	11.1
	2%	DB	11.1	12.0	15.8	21.4	25.1	28.1	31.0	30.6	25.6	19.5	14.2	12.1
		MCWB	8.9	8.5	10.4	13.6	17.3	19.5	19.9	19.5	17.9	15.1	11.6	10.0
	5%	DB	9.5	10.5	13.2	19.2	22.9	25.9	28.8	27.9	22.9	17.8	13.0	10.8
		MCWB	7.7	8.0	9.3	12.5	15.7	18.3	19.1	18.9	16.6	14.3	10.9	8.9
	10%	DB	8.1	9.1	11.4	17.1	20.9	23.6	26.1	25.2	20.8	16.1	11.7	9.2
		MCWB	6.4	6.9	8.1	11.8	14.6	17.0	18.4	17.8	15.7	13.5	9.6	7.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	10.2	12.7	15.8	20.3	21.9	22.2	21.8	19.9	17.0	13.6	11.6
		MCDB	12.2	12.3	16.9	22.3	26.5	29.2	29.7	29.1	26.3	21.4	15.4	13.1
	2%	WB	9.2	9.2	11.0	14.4	17.9	20.1	20.9	20.6	18.5	15.6	12.2	10.4
		MCDB	10.9	11.2	14.4	20.1	23.8	26.4	28.4	28.2	24.3	18.8	13.9	11.9
	5%	WB	7.8	8.1	9.7	13.3	16.4	18.8	19.9	19.5	17.3	14.7	11.0	9.0
		MCDB	9.3	10.2	12.7	18.2	21.7	24.8	26.7	26.1	21.8	17.3	12.9	10.6
	10%	WB	6.4	6.9	8.5	12.0	15.0	17.6	18.9	18.6	16.2	13.7	9.8	7.7
		MCDB	7.8	9.0	11.2	16.5	19.8	22.4	25.1	24.0	19.9	16.0	11.3	9.1
Mean Daily Temperature Range	5% DB	MDBR	4.4	5.5	7.2	9.5	9.8	9.9	9.9	9.8	8.4	6.7	4.7	4.2
		MCDBR	5.3	6.8	10.5	13.3	13.4	13.8	14.4	14.5	11.9	8.3	5.7	4.7
		MCWBR	4.4	4.7	6.6	7.3	7.2	6.9	6.0	6.2	6.1	5.1	4.3	4.0
	5% WB	MCDBR	5.0	5.8	9.2	11.8	12.1	12.3	12.9	12.0	10.4	7.6	5.6	4.8
		MCWBR	4.5	4.5	6.2	6.9	6.9	6.7	5.7	5.8	5.8	5.0	4.5	4.4
Clear-Sky Solar Irradiance	taub	0.320	0.342	0.377	0.412	0.414	0.413	0.427	0.420	0.389	0.358	0.339	0.316	
	taud	2.360	2.342	2.263	2.208	2.239	2.262	2.234	2.270	2.348	2.414	2.384	2.353	
	Ebn at Noon	672	765	805	820	838	841	820	804	787	738	640	606	
	Edh at Noon	61	82	107	128	130	128	130	119	98	76	60	52	
All-Sky Solar Radiation	RadAvg	0.64	1.33	2.40	3.93	4.80	5.14	4.92	4.14	3.01	1.73	0.78	0.50	
	RadStd	0.07	0.20	0.28	0.47	0.52	0.41	0.65	0.33	0.38	0.17	0.09	0.05	